are large benefits in international investments despite correlations changing through time.[14]

2. Exchange rate risk

   For a domestic investor, foreign equities bring exposure to international stock markets as well as foreign exchange rates.[15] Exchange rate movements are an additional source of risk, which may cause domestic investors to shun foreign assets. This explanation was rejected in an early study by Cooper and Kaplanis (1994). More recently, Fidora, Fratzscher, and Thimann (2007) show that exchange rate volatility can explain some, but far from all, of the home bias puzzle.

3. Transaction costs

   Transaction costs for investing abroad have fallen over the last thirty years, and the degree of home bias has also fallen over this time. Ahearne, Griever, and Warnock (2004) compute a home bias ratio, which is equal to one minus the ratio of the share of foreign equities held by U.S. residents to the share of foreign equities in the world portfolio. The home bias ratio is equal to zero when there is no home bias and equal to one when domestic investors are totally home biased. Ahearne, Griever, and Warnock show that the home bias ratio dropped from almost 1.0 in the early 1980s to below 0.8 in the early 2000s.

   We should, however, be skeptical of transaction costs explanations. The lucky U.S. investor does not need to invest directly in foreign markets—foreign companies have long listed their shares on U.S. exchanges through American Depositary Receipts (ADRs) where transaction costs are low.[16] And there are many financial intermediaries, like mutual funds, specializing in international assets, which U.S. investors can easily access. Viewed more broadly, transaction costs could also include the cost of becoming informed about the benefits of diversification. Both actual transaction costs and the costs of becoming financially literate are out of whack with the costs that economic models predict are necessary to match the pronounced home bias of investors. Glassman and Riddick (2001), for example, estimate that transactions costs need to be extraordinarily high—more than 12% per year—to explain observed home bias.

[14] See also Asness, Israelov, and Liew (2011). Chua, Lai, and Lewis (2010), among others, argue that international diversification benefits have decreased over time hand in hand with greater integration, but this is offset to some extent by the entrance of new international vehicles like emerging markets in the 1980s and frontier markets in the 2000s.

[15] Technically it is movements in real exchange rates that matter. If purchasing power parity (PPP) held, then there would be no real exchange rate risk (see Adler and Dumas (1983)). There is a very active literature on PPP. PPP certainly does not hold in the short run, and academics have not reached consensus on whether PPP holds in the long run (twenty- to one-hundred-year horizons). Taylor and Taylor (2004) present a summary.

[16] See Errunza, Hogan, and Hung (1999).